

WINONA, MN, USA

WMO: 726588

Lat: 44.077N

Lon: 91.708W

Elev: 200

StdP: 98.95

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04956

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-20.1	-27.8	0.3	-20.5	-25.4	0.4	-18.8	9.8	-7.4	8.6	-6.9	1.7	300	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.6	23.0	31.4	22.7	28.9	21.4	25.1	30.7	23.8	28.9	22.8	27.4	3.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.2	27.8	22.4	17.5	27.0	21.3	16.3	25.8	77.1	30.8	72.3	29.0	68.1	27.6	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.3	6.3	DB	-27.2	35.9	4.4	3.5	-30.3	38.4	-32.9	40.5	-35.3	42.5	-38.5	45.1	
			WB	-27.3	26.9	4.2	1.5	-30.3	27.9	-32.7	28.8	-35.1	29.6	-38.1	30.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-7.5	-5.2	1.5	8.6	15.0	20.5	22.9	21.5	17.6	10.3	2.7	-4.2
	DBStd	11.77	6.88	6.72	6.35	5.39	4.58	3.72	3.25	3.21	4.50	5.09	5.33	6.13
	HDD10.0	2064	543	425	273	88	9	0	0	0	3	59	226	439
	HDD18.3	3918	801	658	522	294	124	19	3	7	67	254	470	698
	CDD10.0	1595	0	0	9	46	163	316	399	358	229	68	7	1
	CDD18.3	406	0	0	1	2	20	84	143	106	44	6	0	0
	CDH23.3	3562	0	0	2	38	208	751	1284	859	366	54	1	0
	CDH26.7	1094	0	0	0	8	57	231	449	242	97	9	0	0
Wind	WSAvg	2.4	2.6	2.7	2.7	3.1	2.6	2.2	1.9	1.7	2.0	2.5	2.7	2.5
Precipitation	PrecAvg	880	28	23	49	97	102	120	109	124	93	54	51	32
	PrecMax	1086	72	66	85	198	252	236	225	328	255	167	170	63
	PrecMin	622	9	5	9	28	41	47	33	26	22	17	4	7
	PrecStd	134	17	15	22	54	44	57	48	65	57	35	37	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	12.6	22.3	27.4	32.2	34.0	36.0	34.1	32.4	27.6	21.1	12.4
		MCWB	5.3	8.2	13.7	16.1	19.4	23.8	24.3	24.9	22.4	19.1	13.8	9.7
	2%	DB	5.0	9.0	17.4	23.0	27.7	32.1	32.8	32.0	29.1	24.8	16.3	7.9
		MCWB	2.2	5.7	11.6	13.3	17.6	22.5	23.5	23.7	21.3	17.6	11.0	5.4
	5%	DB	2.6	6.0	13.8	21.1	25.9	29.1	32.0	29.0	27.3	22.1	12.9	5.2
		MCWB	1.2	3.0	9.4	12.4	16.9	20.9	23.4	22.2	19.9	15.7	8.5	2.8
	10%	DB	1.3	3.0	11.1	17.4	22.9	27.4	28.8	27.5	25.2	18.7	11.1	2.8
		MCWB	0.3	1.1	7.4	9.7	15.4	19.4	22.0	20.8	18.8	14.1	7.5	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	9.0	15.5	17.2	22.3	25.3	27.3	26.9	24.1	21.7	14.8	10.6
		MCDB	9.0	11.7	21.2	24.4	28.2	32.1	32.1	30.9	29.9	25.8	19.2	12.5
	2%	WB	2.7	5.8	12.7	14.9	20.0	23.7	25.5	24.6	22.5	18.4	11.8	5.6
		MCDB	4.6	9.2	16.2	21.6	25.4	29.5	31.2	29.3	27.4	21.9	15.4	7.6
	5%	WB	1.5	3.4	9.9	13.1	18.6	22.3	24.1	23.4	21.5	16.2	9.5	3.1
		MCDB	2.6	5.6	13.8	19.0	23.5	27.5	29.2	27.6	25.5	21.1	12.9	4.9
	10%	WB	0.4	1.5	6.9	11.2	17.1	21.1	22.9	22.3	20.2	14.2	7.6	1.6
		MCDB	1.3	3.2	10.5	16.8	21.7	25.5	27.7	26.1	24.0	18.2	10.3	2.9
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	9.4	10.7	11.1	10.8	10.6	10.4	11.0	10.1	7.9	7.3
		MCDBR	8.8	10.7	13.5	15.9	14.4	13.0	12.2	12.1	13.0	14.0	12.2	9.4
		MCWBR	7.3	7.9	8.2	8.1	6.6	5.8	5.4	5.5	6.3	7.8	8.0	7.4
	5% WB	MCDBR	8.2	10.0	12.6	13.4	11.7	11.0	10.8	10.3	11.0	11.9	11.3	8.8
		MCWBR	7.0	7.6	8.3	7.9	6.5	5.8	5.6	5.6	6.1	7.8	8.2	7.2
Clear-Sky Solar Irradiance	taub	0.265	0.297	0.325	0.370	0.403	0.410	0.411	0.406	0.375	0.330	0.295	0.268	
	taud	2.395	2.306	2.373	2.315	2.270	2.289	2.292	2.317	2.381	2.496	2.506	2.442	
	Ebn at Noon	872	893	913	893	869	862	855	847	848	850	832	835	
	Edh at Noon	74	98	105	122	131	130	128	120	104	81	68	65	
All-Sky Solar Radiation	RadAvg	1.77	2.84	3.87	4.54	5.35	5.96	6.20	5.24	4.17	2.67	1.74	1.28	
	RadStd	0.18	0.28	0.42	0.46	0.41	0.46	0.26	0.35	0.30	0.36	0.15	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page